Continuous Internal Assessment - 1

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45

1) Online Railway Ticket Booking System:

a) TCP/IP, DNS and URL:

TCP ensures reliable data transfer between the browser and server.

IP identifies the devices using IP addresses.

DNS: converts the website's domain name into its IP address.

URL: Identifies the resources requested by the browser.

It contains the protocol, domain name, path.

Eg: https://railway.com/trains.

b) HTTP Request and Response:

An HTTP Request contains:

* Request Line: Method, URL and HTTP version.
* Headers: Information such as content type and cookies
* Body: Data sent to the server such as booking details.

An HTTP Response contains:

* Status line: HTTP version and status code
* Headers: Content type and caching information
* Body: The requested webpage.

c) HTTP Methods:

Method	Purpose	Idempotent
GET	Retrieve train details	yes
POST	Create a new booking	No
PUT	Replace or update a booking.	yes
DELETE	Cancel a Booking	yes

d) HTTP Status codes:

Code

200 . OK - Request Successful.

201 - Created - Booking Created.

304 - Not Modified - Use cached data.

401 - Unauthorized - Login Required.

404 - Not Found - Resource unavailable.

Stateless HTTP. Cookies, Sessions and HTTPS / TLS

* HTTP is stateless because each request is independent.
* Cookies and sessions maintain user login & booking information.
* HTTP uses TLS to encrypt communication and protect sensitive information.

Problems and Solution:

1. Duplicate Bookings: occur when refreshing resubmits a POST Request.

2. Stale seat availability: * Occurs due to cached responses.
   * Use Cache-Control: no store for live seat data.

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2) Online Course Registration Page:

a) Document Structure and Semantic Elements:

XHTML is a stricter version of HTML

XML Rules

<!DOCTYPE html> declares the document page
<html> is the root element
<head> contains metadata and page title
<meta> provides character encoding.
<body> contains visible content

Semantic Elements

<header> - Page Heading

<nav> - navigation links

<main> - Main content

<section> - Related content

<footer> - Footer Information.

b) Table Elements:

<table> → creates a table

<caption> → provides a table title

<thead> → Contains headings

<tbody> → contains data.

<td> → defines a data cell.

c) Form Elements & Attributes:

action Specifies where form data is sent.

method Specifies GET or POST.

<label> identifies a form field.

<input> Collects text, email, data and other data.

<textarea> collects comments

Pattern → Specifies an input format.

Hyperlinks, Images and URLS:

 Courses → Hyperlink.

<img src = "courses.jpg" alt = "Course Image">

e) HTML and XHTML Rules

HTML	XHTML
Some tags can be unclosed	Every tag must be closed
Attribute values may be unquoted	Attribute value must be quoted
Tags are generally case-insensitive	Tags must be lowercase.
Improperly nested	Properly Nested.

3) XML and CSS

a) XML Document for a library Catalogue.

XML stores and exchange structured data.

XML Document:

```
<? xml version = "1.0" encoding = " UTF - 8"?>
<library xmlns = " http. //example. com / library ">
    <book ISBN = "978001">
    <title > Internent programming </title>
    <author> James Gosling </author>
    <publisher> ABC Publishers </ publishes>
    <year> 2084 </year>
    <price> 450 </ price>
  </book>
  <book ISBN = "978003">
     < title > web Technology </title>
     <year> 2024 </ year>
     <price> 600 </ price>
  </book>
</library>.
```

* DTD defines the allowed elements and attributes
* XML schema provides stronger validation, including data types
* Three - well formedness errors:
    1. Missing closing Tags
    2. Unquoted attribute values
    3. Incorrect nesting of elements.

XML Stores & exchanges structured data. JSON is light weight data - exchange format.

CSS Cascade and Selectors:

Given:

HTML

```
<p id="alert" class="notice"
   style="color: orange;">Library closed </p>
```

CSS

```
p{color: blue}
.notice{color: green;}
#alert{color: red;}
```

If !important is added to .notice, green will be applied because it overrides normal inline styles.

* The cascade considers importance, specificity and source order.
* Inheritance passes certain properties such as colour, from parent to child elements

CSS Types:

Inline: Written inside style attributes

Internal: Written inside <style> element.

External: Written in separate .css file.

Selectors.

* Element: p
* Class · notice
* ID: #alert
* Child: div>p
* Pseudo-class: p: hover.

4. Responsive News Portal

a) Box Model and Margin Collapsing

The css model consist of content, padding, border and margin.

```
* {
    box-sizing border-box;
}
.container {
    margin: 20px;
    Padding: 15px;
    border: 1px solid black;
}
```

Border-box width includes content, Padding & border

Margin collapsing occurs when vertical margins of adja blocks

display and Positioning:

* Static: Default Positioning
* Relative: Positioned Relative to its normal position.
* Absolute: Positioned Relative to its nearest ancestor.
* Fixed: Remains Fixed. relatives to the viewport.
* Sticky: Behaves normally until a scroll position.

```
nav {
    position: sticky;
    top: 0;
    background: white;
}
```

c) Flexbox and css grid:

Flex-box is used for one-dimensional layout such as navigation bars.

```
nav {
    display: flex
    justify context: space-between;
    align items: center;
}
```

CSS grid is used for two-dimensional layouts

```
.articles {
    display: grid;
```

```
    grid-template: (3, 1fr);
}
```

d) Responsive Design:

rem: Relative to the root font size
em: Relative to the current or font size
vw: Relative to viewport width
vh: Relative to viewport height.

```
@media (max-width: 600px) {
    .articles {
        grid-template-columns;
    }
    .sidebar {
        display: none;
    }
}
```

e) Typography, Colours & CSS variables

1. Use readable font sizes.
2. Maintain proper colour contrast
3. Use consistent colours
4. CSS variables helps reduce style.

```
root {
    -- primary: #333;
    -- background: #f5f5f5;
}
```

Flexbox, grid and media queries create a flexible website that works on all screen sizes

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5. Explain Js and Dom manipulation in online bookstore:

) Java script fundamentals:

Variables:

* let → value can change
* Const → cannot be reassigned.

```
let quantity = 2;
Const price = 500;
let books = ["IP", "Java"];
let book = {
        name: "IP",
        Price: 500
};
```

b) Dom selection and Manipulation:

getElementById() → Selects element by ID
querySelector() → Selects first matching element
classList → Add or removes classes.

```
let msg = document.getElementById("message");
msg.classList.add("success");
```

) Event Handling:

Click
Input
Change
Submit
Event Bugging
Event Delegation.

Form validation.

Regular expression are used to validate user inputs

It checks:

① Minimum 8 characters.
② Atleast one uppercase letter.
③ Atleast one digit.

Fetching Data Asynchronously:

* Fetch API retrieves data from a server without reloading the webpage.
* Use textContent to display user data safely
* Display proper error messages
* Reduces event listeners.
* Allows users to retry
* Preserve cart information.